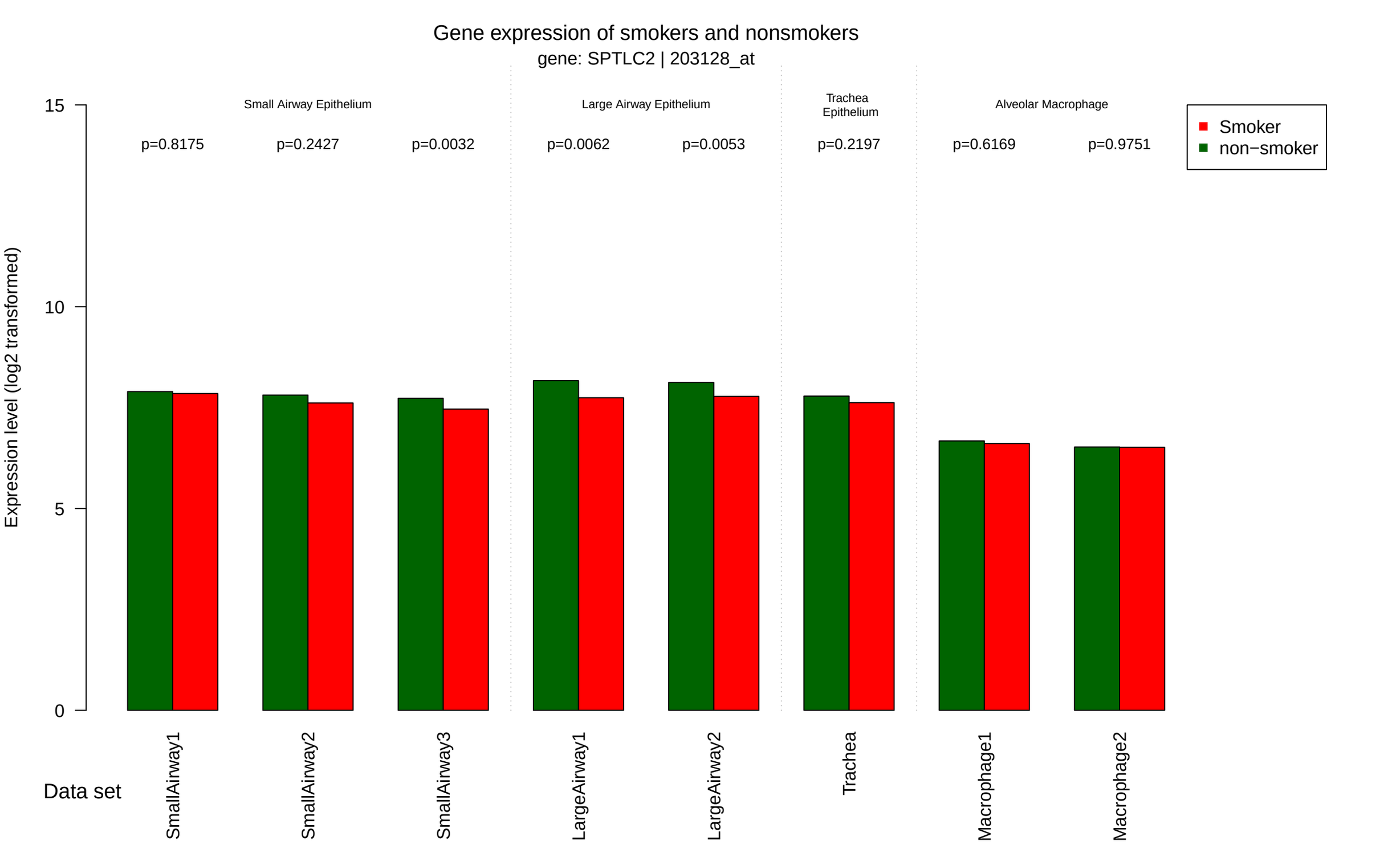
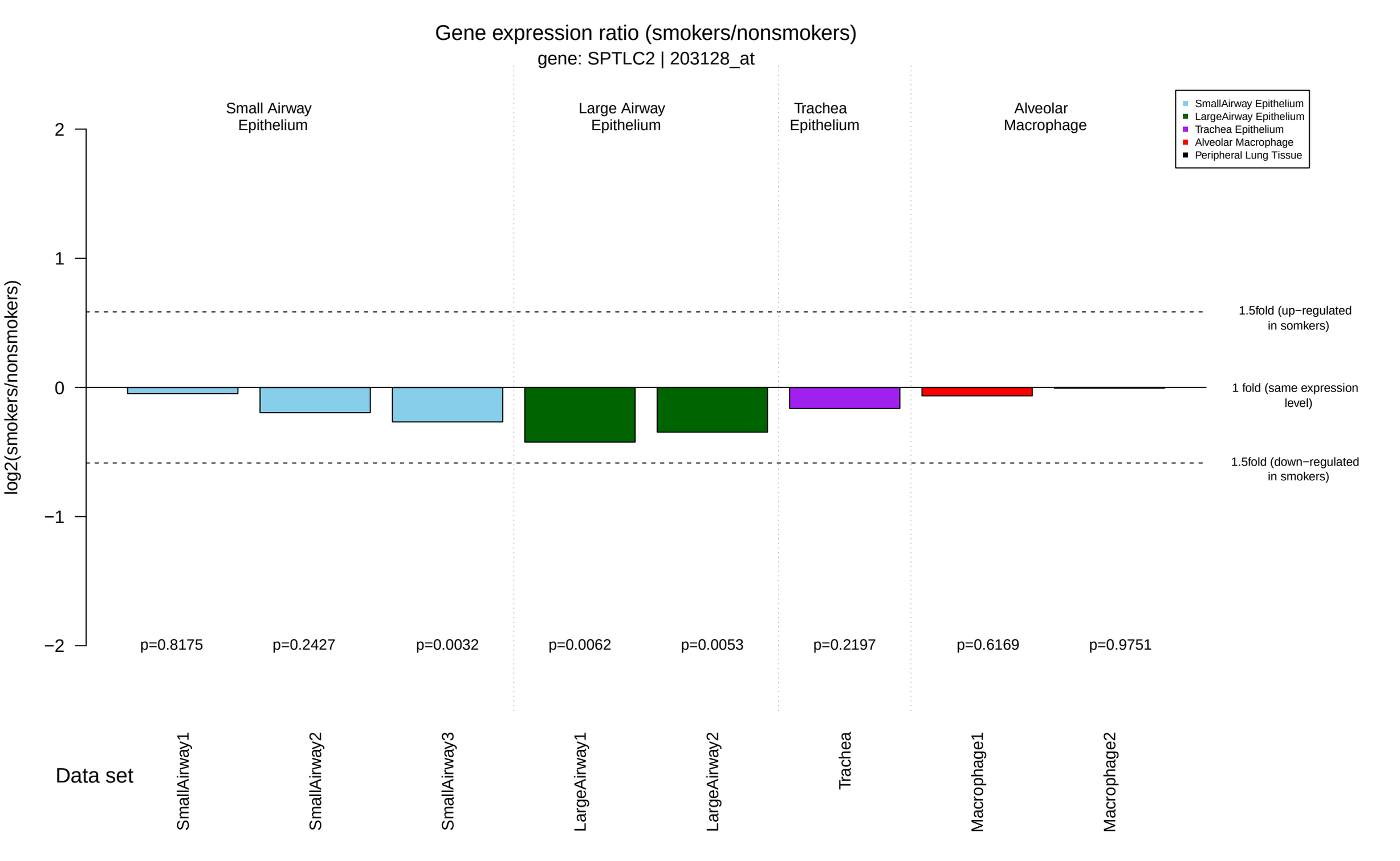
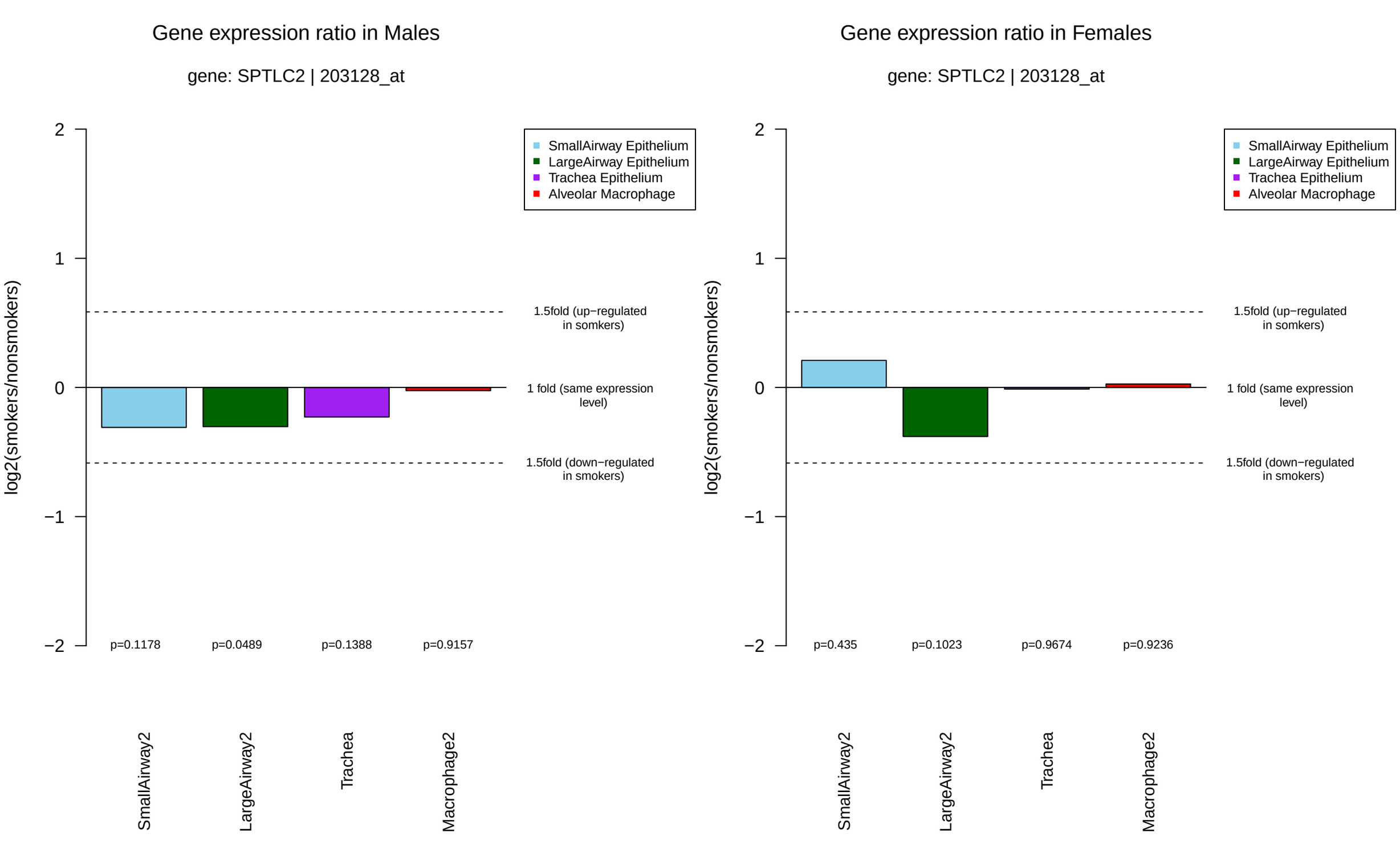
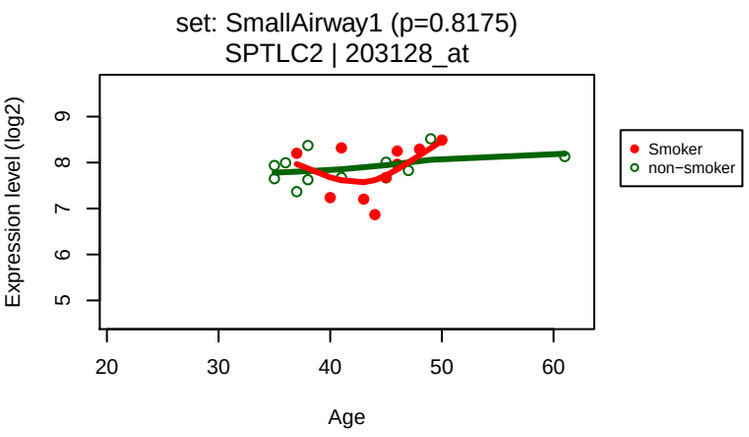
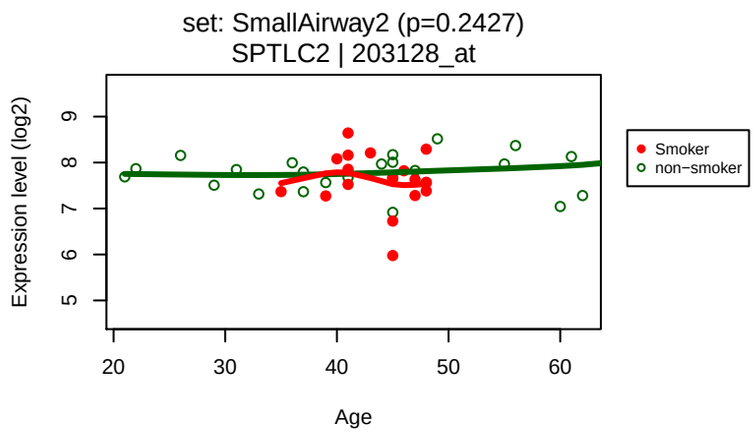
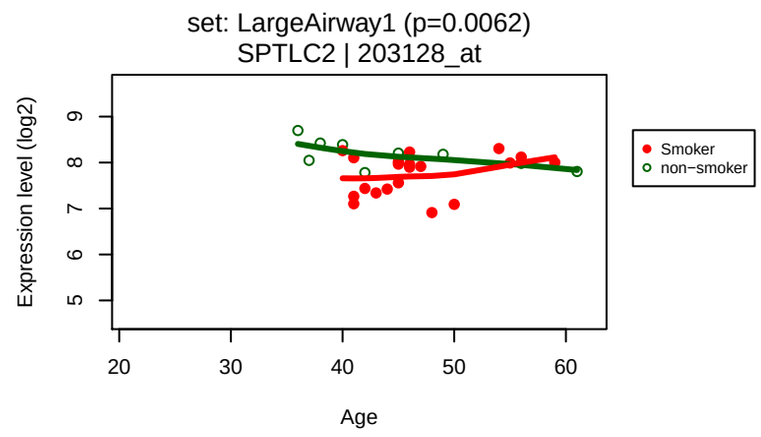

Gene expression of smokers and nonsmokers
gene: SPTLC2 | 203128_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8175
p=0.2427
p=0.0032
p=0.0062
p=0.0053
p=0.2197
p=0.6169
p=0.9751
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

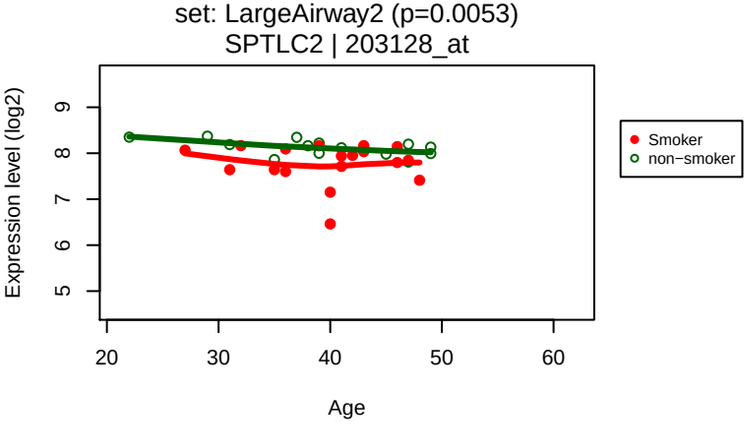

Gene expression ratio (smokers/nonsmokers)
gene: SPTLC2 | 203128_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8175
p=0.2427
p=0.0032
p=0.0062
p=0.0053
p=0.2197
p=0.6169
p=0.9751
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

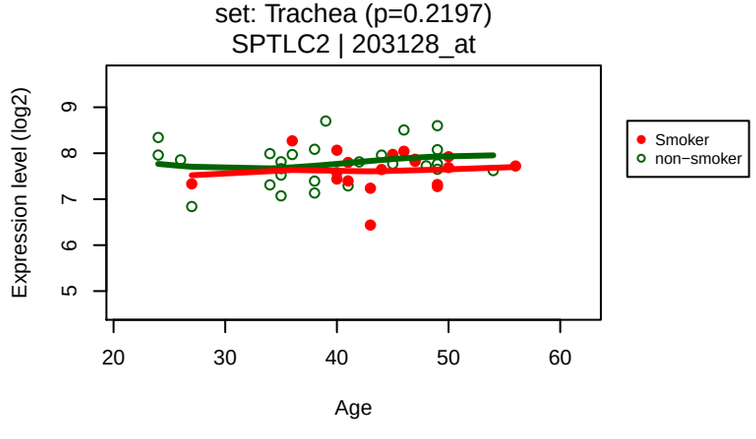

Gene expression ratio in Males
gene: SPTLC2 | 203128_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1178
p=0.0489
p=0.1388
p=0.9157
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPTLC2 | 203128_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.435
p=0.1023
p=0.9674
p=0.9236
SmallAirway2
LargeAirway2
Trachea
Macrophage2

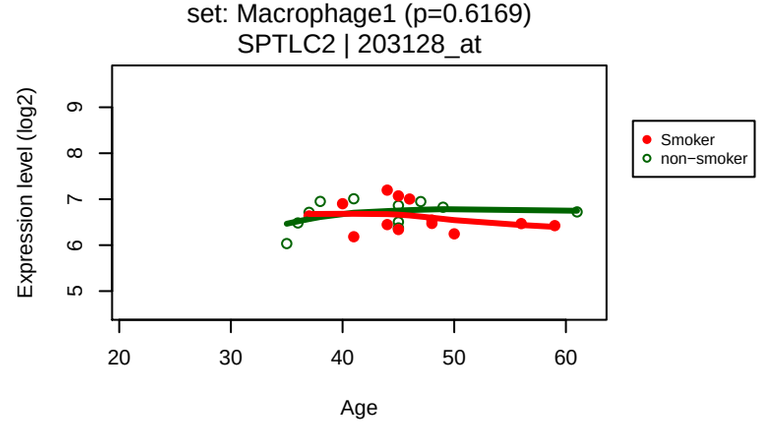

set: SmallAirway1 (p=0.8175)
SPTLC2 | 203128_at
Expression level (log2)
Age
Smoker
non−smoker

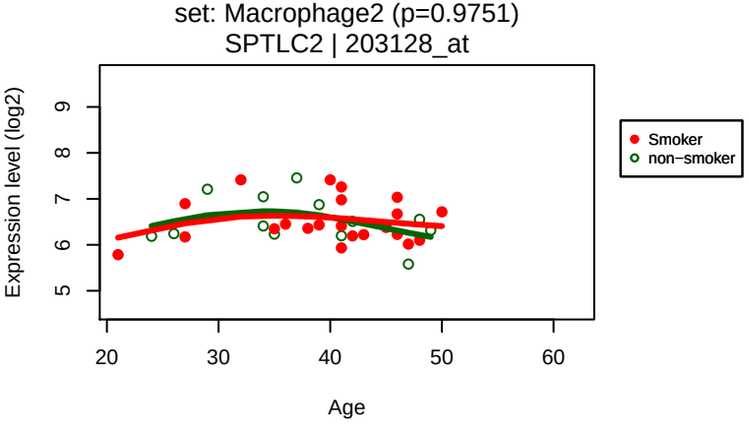

set: SmallAirway2 (p=0.2427)
SPTLC2 | 203128_at
Expression level (log2)
Age
Smoker
non−smoker

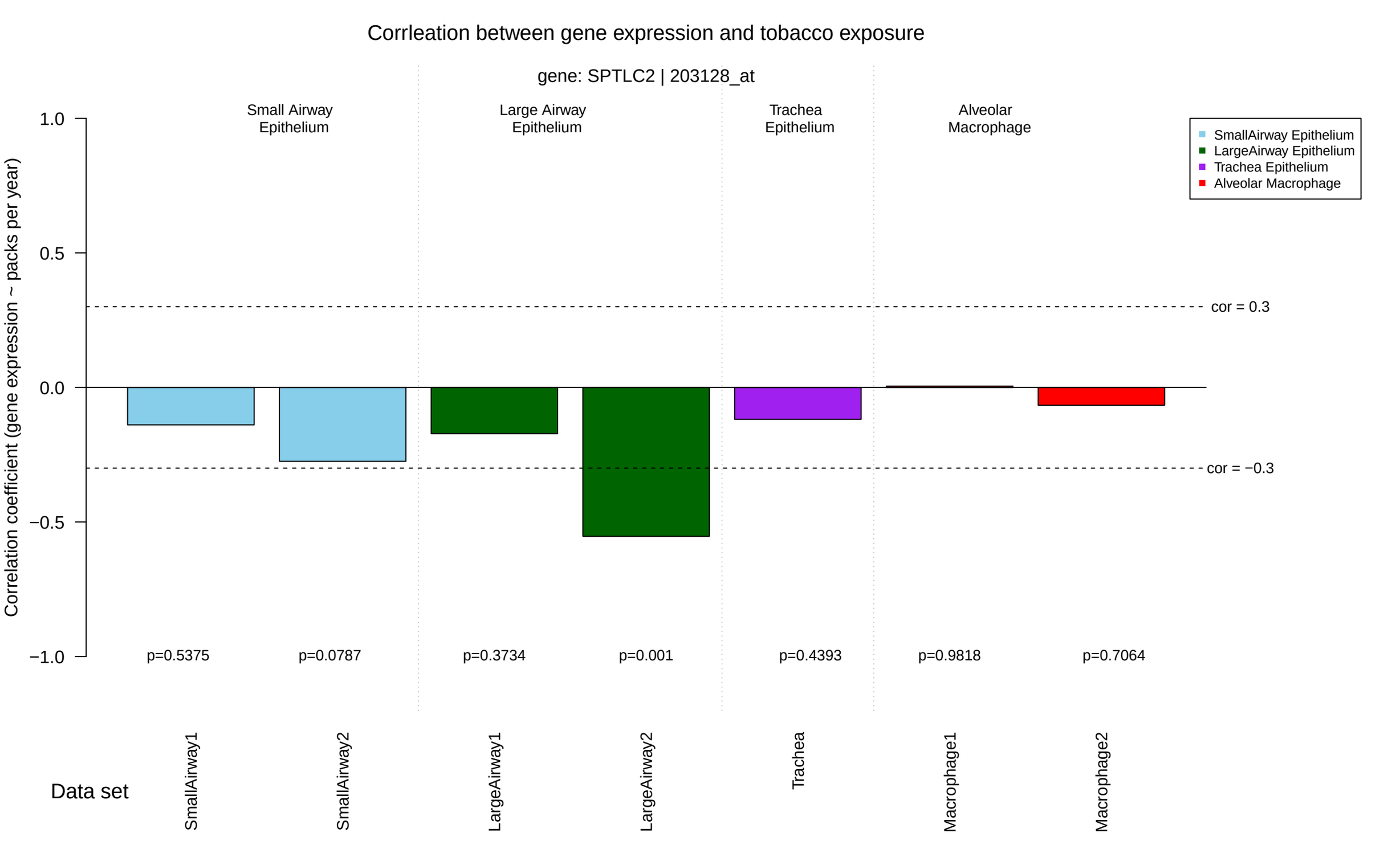

set: LargeAirway1 (p=0.0062)
SPTLC2 | 203128_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0053)
SPTLC2 | 203128_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2197)
SPTLC2 | 203128_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6169)
SPTLC2 | 203128_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9751)
SPTLC2 | 203128_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPTLC2 | 203128_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5375
p=0.0787
p=0.3734
p=0.001
p=0.4393
p=0.9818
p=0.7064
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2